TECHNICAL EVALUATION REPORT

Selection Functionals on Path Space for Stochastic Processes Without a Generator: Finite-Grid Gibbs Selection and Self-Normalized Estimation

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1 Summary of the submission

The paper develops an estimation theory on finite-grid path space for systems whose two-time transition families violate the Chapman–Kolmogorov (CK) factorisation and therefore admit no generator on the observed state space reproducing them.

The construction is as follows. A proposal mechanism Π_M supplies i.i.d. candidate paths $\xi = (\xi_0, \dots, \xi_M)$ on a fixed grid. A surrogate drift–diffusion pair (b_ν, σ_ν), explicitly not the truth, defines the discrete Onsager–Machlup (OM) action $S_{\nu,M}[\xi]$, the Euler–Maruyama discretisation of the classical functional. The selection weight is $w_\nu = e^{-\lambda S_{\nu,M}}$, the selected measure is the Gibbs reweighting $dQ_{\nu,M} = w_\nu \, d\Pi_M / Z_\nu$, and the estimator is the self-normalised importance-sampling (SNIS) ratio. Four groups of results follow: consistency, a central limit theorem and asymptotic mean-squared error (§4); two-tier finite-sample concentration, polynomial under finite chi-square and exponential under bounded weights (§5); variance and effective-sample-size diagnostics organised by the chi-square divergence (§6); and a conditional transfer to the target process under an explicit total-variation hypothesis, flanked by an obstruction bounding every divisible surrogate away from an indivisible target in kernel norm (§7). Three surrogate families are characterised (§8), one example is worked in closed form with a numerical study (§9), and a structural comparison with misspecified-model theory, path-space variational inference and generalised Bayes closes the argument (§10).

The substantive standard applicable here is reproducibility, and the paper's central methodological claim is a mathematical one. The paper is therefore judged it on three axes: whether the theorems are correct as stated and correctly scoped; whether the worked example reproduces independently; and whether the operational role claimed for the action is new and correctly positioned.

2 Overall assessment

Two things should be said before any criticism, because they are the dominant impression the manuscript leaves and they bear directly on the standard being applied.

First, the paper is committed with reproducibility standard at a high level: The worked example of §9 from the printed specification (proposal, surrogate, action, observable, discretiser, grid) was implemented and all reported quantity could be recomputed independently. All fourteen agree to the precision printed, including the four-grid refinement table and the exponential growth rate of the chi-square divergence. The verification record below gives the comparison in full. The paper ensures full transparency in this regards; the inclusion of clear closed forms allows complete verification of the example without any code.

Second, the manuscript is notable for its rigorous delineation of scope and explicit identification of theoretical limitations. In section §2, the author demonstrates that their natural continuous-path formulation is degenerate, which properly justifies the transition to a finite-grid setting. Remark 4.2 correctly avoids an invalid finite-N mean-squared-error identity and rejects a common factorization; independent verification confirms that this standard factorization overstates the variance of the worked example by a factor of 2.2. Furthermore, Remark 5.4 transparently notes that the proposed Tier II bounds are vacuous within the paper's own example. Remark 2.2 distinguishes between three exclusions often conflated in the literature, accurately citing Feller regarding the failure of the converse implication. Finally, sections §7 and §10.4 properly withdraw an unconditional bound on selection bias and an unsustainable structural distinction, respectively, providing sound mathematical justifications for both. Each of these stated limitations and corrections was verified

and found to be accurate

The results presented in theorems are correct to my judge.

Three defects bear on the substance of the argument. None invalidates a theorem; all three concern what the theorems are said to mean.

First, the inference that opens §7.2 is invalid, and the manuscript contradicts it four paragraphs later (Finding 3). The section asserts that the conditional structure of the transfer theorem is "forced, not chosen", because no divisible surrogate drives the selection bias to zero. It does not follow. The selected measure is a Gibbs tilt of the *proposal*, not the law of the surrogate, and the proposal is a free object; kernel-norm separation of the surrogate and total-variation separation of the selected measure are logically independent. One can settle the point by construction: indeed, by exhibiting a grid, an indivisible target, a divisible Ornstein–Uhlenbeck surrogate and a proposal for which the selected measure equals the true grid law *exactly*, so the selection bias is zero while the obstruction of §7.2 applies in full force.

Second, the selection weight is, up to a constant, the surrogate's own path density, and the selected measure is therefore a product of two path densities rather than a ratio (Finding 2). Multiplying path densities adds their precisions: in the worked example the selected measure is an Ornstein–Uhlenbeck-type chain with $\sigma_{\mathrm{eff}}^{-2} = \sigma_{\Pi}^{-2} + \lambda\sigma_{\nu}^{-2}$, confirmed numerically. This is the structural fact behind everything distinctive in the paper's behaviour (the chi-square blow-up under refinement, the choice of proposal needed for the transfer hypothesis to be attainable, and the failure of the temperature schedule proposed for a continuum theory) and it is nowhere stated.

Third, the claimed third operational role for the action is not positioned against the path-reweighting literature (Finding 8), in which reweighting an externally generated path ensemble by a path action is established practice. The distinction that does survive is sharper than the one claimed and follows from the second finding: that literature reweights by a *difference* of actions, and this construction by a *single* action.

Six further qualifications are recorded below, of which the most consequential are that the worked example never exercises the Itô-correction term (Finding 6); that the convergence study's variance narrative is not estimable at fifty replications (Finding 7); and that the calibrated parameter is called optimal for an objective it does not optimise (Finding 5).

3 Section-by-section analysis

3.1 The classical action and its two breakdowns (§2)

Proposition 2.1 is correct and its proof is correct: an Itô diffusion is Markov, the two-time kernels of a Markov process satisfy CK, so a drift–diffusion pair reproducing an indivisible family would force the deficit to vanish. I note in the paper's favour that the direction of the Markov–CK implication is used correctly here, and that Remark 2.2 states the failure of the converse with its standard reference. This is the point at which comparable arguments in the wider literature go wrong, and the manuscript does not.

Remark 2.2 is the most valuable paragraph in the section. It separates (E1) no generator on the observed state space, (E2) no finite hidden-Markov enlargement, (E3) no Markovian enlargement of any kind; observes that only (E1) follows from CK failure; and records that CK failure does not exclude a path-space law either. The scoping is correct and the concession that an earlier statement overstated the exclusion is made without hedging.

The divergence term is not the Itô term of the Girsanov derivative. After equation (4) the manuscript describes $\nabla \cdot b$ as "the Itô correction from the Girsanov density of μ_X relative to the law of $x_0 + \sigma W$". The Girsanov Radon–Nikodym derivative in Itô form, $\exp\big(\int \sigma^{-2} b \cdot dW - \frac{1}{2}\int |\sigma^{-1} b|^2\, dt\big)$, does not display a divergence term at all. The divergence enters at the next step: in the Dürr–Bach small-ball passage through the tube Jacobian, and (in the special case of a gradient drift $b = -\nabla V$) through Itô's formula applied to V in converting the stochastic integral, which is where the name "Itô correction" comes from. As written the attribution is compressed to the point of being wrong for general drift. This is a matter of scholarly precision rather than a failure of any theorem, but it is the one place where the manuscript's account of a classical object is inaccurate. **[R]**

Finding 1 (The degeneracy is of the pointwise weighting, not of continuous-path inference)**.** Proposition 2.3 is correct in all four cases. The Brownian case is standard; the bridge argument (subtracting a random but almost surely linear path preserves absolute continuity in both directions) is valid; the Ornstein–Uhlenbeck argument via the Doob representation and a smooth strictly increasing time change is valid; and the Girsanov case follows from shift-invariance of the absolutely continuous set together with equivalence of laws. I record that the proposition is a self-directed negative result: it establishes that the natural continuous-path version of the author's own construction does not exist, and the paper builds on that rather than around it.

What is overstated is the reading placed on it. The manuscript writes that "the continuous-path formulation of the selection functional is degenerate" and that Proposition 2.3 "invalidates the continuous-path formulation as written". What the proposition invalidates is the definition (6): the convention $S_\nu^{\mathrm{OM}}[\xi] = +\infty$ off the absolutely continuous paths, evaluated pointwise on raw diffusion samples. That convention is not how the classical object is used. The Onsager–Machlup functional is a generalised small-ball density evaluated on *regular reference* paths (the manuscript's own equation (4) is a ratio of tube probabilities around two smooth reference paths) and it is not a density to be evaluated at a typical Brownian sample. A Girsanov change of path measure, likewise, is defined directly on path space and assigns no action to any individual sampled path.

This is not merely a matter of phrasing, and the verification record below settles it numerically: a *ratio* weight on the same grids, which is what path-space reweighting actually uses, has chi-square divergence 0.1645, 0.1601, 0.1580, 0.1569, 0.1564, 0.1562 at $M = 25, \ldots, 800$ (bounded, and converging to a finite continuum value) against the selection weight's divergence to 9.8×10^{49} over the same range. Continuous-path reweighting is not obstructed; the pointwise Gibbs weighting is. Every occurrence of "the continuous-path formulation is degenerate" should be narrowed to the formulation defined in (6). **[R]**

3.2 The grid construction (§3)

Finding 2 (The weight is the surrogate's path density, and the selected measure is a product)**.** The discrete action (7) has kinetic sum $\frac{1}{2}\sum_m |\sigma_\nu^{-1}(\xi_{m+1} - \xi_m - b_\nu(\xi_m)\delta t)|^2/\delta t$ and divergence sum $D[\xi] = \frac{1}{2}\sum_m (\nabla \cdot b_\nu)(\xi_m)\,\delta t$. The first is exactly the exponent of the Euler–Maruyama transition density p_ν of the surrogate, so in general

$$e^{-\lambda S_{\nu,M}[\xi]} \;=\; (2\pi\sigma_\nu^2 \delta t)^{\lambda d M/2}\, p_\nu(\xi)^\lambda\, e^{-\lambda D[\xi]},$$

and whenever $\nabla \cdot b_\nu$ is constant (which covers the Ornstein–Uhlenbeck family and the whole of the worked example) the last factor is a constant and

$$Q_{\nu,M} \;\propto\; \Pi_M \cdot p_\nu^\lambda.$$

The selection weight is not merely "action-shaped"; it *is* the surrogate's own path density, tempered by λ. What distinguishes the construction from the path-reweighting literature is accordingly not that the weight fails to be a density, but that it enters *multiplicatively* where reweighting methods divide.

This identification is not a restatement. It fixes the selected measure's diffusion coefficient, and with it every degeneracy the paper reports. For Gaussian proposal and surrogate, multiplying densities adds precisions: in the notation of §9, $\Sigma_\lambda = (K_\Pi + \lambda K_\nu)^{-1}$, whose interior rows correspond to an Ornstein–Uhlenbeck-type chain with

$$\frac{1}{\sigma_{\text{eff}}^2} \;=\; \frac{1}{\sigma_\Pi^2} + \frac{\lambda}{\sigma_\nu^2}. \tag{1}$$

The selected measure and the proposal thus have *different quadratic variation*, and diffusion measures with different quadratic variation are mutually singular. That is the mechanism behind the chi-square blow-up reported in §9.4, and it is more informative than the account given there, which attributes the blow-up to the continuum degeneracy of Proposition 2.3.

Identity (1) is also the constructive half of the theory, and it is what Finding 3 exploits: it says exactly how the proposal must be chosen if the selected measure is to land on a target with a prescribed diffusion coefficient. The manuscript should state the identification, derive (1), and say plainly that λ is not only a temperature but a diffusion-coefficient control. **[M]**

Lemma 3.6 is correct: under $\nabla \cdot b_\nu \geq -c_\nabla$ the divergence sum is at least $-c_\nabla T/2$, giving $W_{\max} = e^{\lambda c_\nabla T/2}$, and the chi-square bound follows from $\mathbb{E}[w^2] \leq W_{\max}^2$. The Ornstein–Uhlenbeck specialisation $W_{\max} = e^{\lambda k T d/2}$ is right, and equals e in the worked example. The lemma delivers existence and says nothing about conditioning, and the manuscript is careful to keep the two apart: the same example has bounded weights and $W_{\max}/Z_\nu \approx 3.5 \times 10^7$, so the selected measure is mathematically proper while the importance sampling is severely degenerate. The claim that condition (P3) is the minimal requirement for finite asymptotic variance uniformly over bounded observables is correct in both directions, and its position strictly between finite Kullback–Leibler divergence and uniform boundedness of the ratio is correct.

Remark 3.9, the causal factorisation of the weight, is correct and follows immediately from the adaptedness of the kinetic and divergence sums.

3.3 Theorems A, B and C (§4–§6)

Theorem 4.1. Correct. Part (i) is the strong law applied to numerator and denominator with the continuous mapping theorem; part (ii) is the standard delta-method or Slutsky argument with the usual SNIS asymptotic variance; part (iii) is the upgrade of convergence in distribution under uniform integrability. The hypotheses are exactly what the proofs use. Part (iii) should however be read as the conditional statement it is: no primitive sufficient condition for uniform integrability of $\{N(\hat{F}_N - F^*)^2\}$ is supplied, and in the heavy-tailed-weight regime the paper's own diagnostics describe, uniform integrability is not a formality. Either primitive conditions or an explicit acknowledgement that the asymptotic mean-squared-error statement is unavailable in practice belongs in the statement. **[R]**

Remark 4.2. Correct and worth preserving. Second moments genuinely do not yield a finite-N identity when the denominator is random, and the factorisation $(1 + \chi^2)\,\mathrm{Var}_Q(F)$ genuinely fails; the manuscript quantifies the failure in its own example.

Theorem 5.1. Correct. On $\{B_N \geq Z_\nu/2\}$ one has $|\hat{F}_N - F^*| \leq 2|N^{-1}S_N|/Z_\nu$; Chebyshev gives $4\sigma^2/(Nt^2)$; Cantelli applied to the lower deviation of the denominator gives exactly $4\chi^2/(N + 4\chi^2)$,

relaxed to $4\chi^2/N$, using $\mathrm{Var}(w_\nu) = Z_\nu^2\chi^2$; the union of the two events gives (13). For (15) the split on the denominator event with $|\hat{F}_N| \le F_{\max}$ gives $4\sigma^2/N + 16F_{\max}^2\chi^2/N$ as printed.

Theorem 5.2 and Corollary 5.3. Correct. With $u = Z_\nu t/2$, $b = 2F_{\max}W_{\max}$ and $\tilde{\sigma}^2 = Z_\nu^2\sigma^2$, Bernstein returns exactly the exponent $Nt^2/(8\sigma^2 + \frac{8}{3}F_{\max}W_{\max}t/Z_\nu)$ of (16); the one-sided Hoeffding bound on a variable of range $W_{\max}$ returns $\exp(-NZ_\nu^2/2W_{\max}^2)$; and the Hoeffding variant with summand range $4F_{\max}W_{\max}$ returns the exponent $NZ_\nu^2t^2/(32F_{\max}^2W_{\max}^2)$ of (17).

Remark 5.4. Correct. In the worked example the denominator exponent is $\exp(-4.2\times10^{-16}N)$, so Tier II is vacuous below $N \sim 10^{15}$; the paper says so. The observation deserves emphasis in its own right: the strongest-looking exponential theorem in the paper is the least informative one in the paper's own example.

Theorem 6.1. All three parts correct. The variance identity and $\mathbb{E}_Q[\rho_\nu] = 1+\chi^2$ are immediate; the effective-sample-size limit follows from the strong law, and is a property of the weight sample rather than an observable-specific efficiency identity; the divergence ordering is Jensen, equivalently monotonicity of Rényi divergences in the order.

The two sample-size scales. The manuscript's treatment is careful and faithful. Chatterjee and Diaconis establish a cut-off at $\exp(D_{\mathrm{KL}})$ on the logarithmic scale, sufficient uniformly over an $L^2(Q)$ ball and necessary for the class rather than for each fixed observable; the manuscript states it in exactly that form, which is a more careful reading than the result usually receives. The ordering $\exp(D_{\mathrm{KL}}) \le 1 + \chi^2$ is the paper's own Theorem 6.1(iii).

The diagnostic thresholds are heuristics and should say so. Remark 6.2 recommends treating the estimator as unreliable when $\hat{N}_{\mathrm{eff}} \lesssim 10^2$ and reaching for Pareto-smoothed importance sampling when $\hat{N}_{\mathrm{eff}}/N \lesssim 10^{-2}$. Neither number follows from Theorem 6.1 or from anything else in the paper; they are operational rules of thumb presented in the same voice as the theorems that precede them. The smoothing method invoked carries its own diagnostic (the Pareto shape estimate $\hat{k}$, presented by Vehtari and coauthors as a finite-sample convergence-rate diagnostic) which is the quantity a reader following that recommendation would actually compute. The thresholds should be labelled as heuristics, and the shape diagnostic reported alongside them wherever the smoothing is recommended. **[R]**

3.4 The truth link and the obstruction (§7)

Theorem 7.1. Correct. With $|F| \le F_{\max}$, $|\int F\, d(Q-P)| \le F_{\max}\|Q-P\|_1 = 2F_{\max}d_{\mathrm{TV}}$, and the reduction to terminal marginals for terminal observables is valid. The theorem is deliberately conditional on (D1) and says nothing about how the tolerance is obtained.

Proposition 7.3. Correct. Inserting the divisible family's own factorisation and splitting the composite term as $[\tilde{\Gamma}(t \leftarrow r) - \Gamma(t \leftarrow r)]\tilde{\Gamma}(r \leftarrow s) + \Gamma(t \leftarrow r)[\tilde{\Gamma}(r \leftarrow s) - \Gamma(r \leftarrow s)]$ and using the contraction hypothesis on one factor of each product gives the three-pair bound (19), and the $D/3$ corollary follows. It is a statement about families of *kernels* in operator norm, and the two comments on scope that follow it are both correct and both necessary.

Finding 3 (The conditional structure is not forced, and the manuscript says so itself)**.** Section §7.2 opens: "The conditional structure of Theorem 7.1 is forced, not chosen: no choice of divisible surrogate drives the selection bias to zero, because divisible families are uniformly bounded away from an indivisible target's kernels." Four paragraphs later the same section states that the obstruction "says nothing quantitative about the total-variation distance in (D1), which involves the proposal and the temperature as well". The two statements are inconsistent, and the second is the correct one.

The inference fails for a structural reason. $Q_{\nu,M}$ is not the path law of the surrogate; it is a Gibbs tilt of the proposal, and by Finding 2 the surrogate enters only through the factor p_ν^λ. The proposal is an unconstrained input (Definition 3.1 asks only for a probability measure on X_M with an i.i.d. sampling rule) so it can be chosen to compensate the tilt. Separation of the surrogate's kernels from the target's, which is what Proposition 7.3 bounds, and separation of the selected measure from the target's grid law, which is what (D1) bounds, are therefore logically independent quantities.

The question could be settle by construction along the following line: on a grid with $M = 8$ and $\delta t = 1/8$, take the surrogate to be Ornstein–Uhlenbeck with $k_\nu = 2$ (divisible, with a contraction semigroup) and $\lambda = 1/2$. Take the proposal to be the centred Gaussian with precision $K_\Pi = A(a_\Pi)/\delta t + \varepsilon(e_1 e_8^\top + e_8 e_1^\top + e_2 e_7^\top + e_7 e_2^\top)$, $a_\Pi = 1 - \delta t$, $\varepsilon = 2.5$: a perfectly legitimate proposal, positive definite (least eigenvalue 0.339) and trivially samplable. Take the target grid law to be the centred Gaussian with precision $K_\Pi + \lambda K_\nu$ (least eigenvalue 1.270). Then by construction

$$Q_{\nu,M} \;=\; P_{\mathrm{true},M} \quad \text{exactly}$$

so $d_{\mathrm{TV}} = 0$, hypothesis (D1) holds with $\eta_{\nu,M} = 0$, and the selection bias vanishes identically. The target is nonetheless indivisible: for a centred Gaussian family the two-time kernels compose only if $C_{ts}C_{rr} = C_{tr}C_{rs}$, and here the direct conditional-mean coefficient at the triple $(s, r, t) = (2, 4, 6)$ is -0.008588 against a composed value of 0.305127, a relative violation of 36.5 at the worst triple. Proposition 7.3 therefore applies in full force and places every divisible family at operator-norm distance at least $D/3$ from the target's kernels, while the selection bias is zero.

The required correction is a deletion and a replacement. The obstruction should be presented only as what it is: a reason why *improving the surrogate's kernels* cannot by itself solve the truth-link problem. It gives no lower bound on the selection bias, and the sentence asserting that the conditional structure is forced should go.

The positive content that replaces it comes from Finding 2. Since the tilt shifts the diffusion coefficient by (1), a proposal that is to place the selected measure on a target of diffusion coefficient σ^2, using a surrogate with $\sigma_\nu = \sigma$, must satisfy $\sigma_\Pi^2 = \sigma^2/(1 - \lambda)$ (a proposal *rougher* than the target by a factor that diverges as $\lambda \uparrow 1$, and no finite-diffusion proposal compensates at $\lambda \geq 1$). That is a design rule, it is checkable, and it is the substantive statement the section should be making about when (D1) is within reach. **[M]**

Finding 4 (For the natural class of proposals the transfer hypothesis fails under refinement)**.** Finding 3 shows that (D1) is attainable when the proposal is designed against the tilt. It is not attainable otherwise, and the natural choice is the undesigned one: a proposal is described throughout as a mechanism generating plausible candidate paths (a forward simulator, a neural-SDE generator, a bridge construction) which is to say, a process with the target's own roughness. For any such proposal the tilt moves the selected measure off the target's quadratic variation, and the two separate. Computing the Hellinger affinity ϱ between $Q_{\nu,M}$ and Π_M in the worked example and using $d_{\mathrm{TV}} \geq 1 - \varrho$ gives

M	25	50	100	200	400
ϱ	4.73×10^{-1}	2.27×10^{-1}	5.20×10^{-2}	2.73×10^{-3}	7.57×10^{-6}
$d_{\mathrm{TV}} \geq$	0.527	0.774	0.948	0.9973	0.99999

so a target sharing the proposal's diffusion coefficient is at total-variation distance tending to one from the selected measure, the tolerance $\eta_{\nu,M}$ tends to one with it, and the bound $2F_{\max}\eta_{\nu,M}$ exceeds the range of the observable.

This does not contradict Theorem 7.1, which is a correct conditional statement, nor does it apply at a fixed grid, where the manuscript is careful to remain. What it does is redirect the open problem of Remark 7.2. The obstruction to controlling $\eta_{\nu,M}$ in the untuned case is not the indivisibility of the target but the quadratic-variation shift introduced by the weight, and no invariant of the target family (integrated deficit, liftability profile or memory rank) can control a quantity that depends on the proposal–surrogate pair. Either the open problem should be restated as the design problem of Finding 3, or the construction should be modified so that the selected measure preserves the proposal's quadratic variation. **[M]**

3.5 Surrogate families and calibration (§8)

The three characterisations are correct. The Ornstein–Uhlenbeck case has constant divergence and hence the automatic bound; the polynomial case with $M_p \geq 2$ has divergence unbounded below, as the double-well example $b = x - x^3$ shows, and the paper correctly says the automatic bound is then unavailable and (P3) must be verified. The Mori–Zwanzig discussion is accurate on the point that matters: projection produces memory rather than removing it, and a divisible surrogate is obtained only by a deliberate truncation, which is stated as an additional approximation.

Section §8.1 contains a correction of an earlier reading that deserves notice: the action accumulates over the whole horizon, so the restoring rate controls an effective decay of sensitivity and not locality of the functional. This is correct, and it is consistent with Finding 2: a path density is not a local object.

Finding 5 (The calibrated parameter is called optimal for an objective it does not optimise)**.** Section §8.4 defines $\hat{\nu}$ as the minimiser of the empirical surrogate action and identifies its population limit $\nu^* = \arg\min_\nu \mathbb{E}\, S_{\nu,M}[\xi_{\text{obs}}]$ as "the best reweighting parameter within the chosen family". The regularity discussion around it is careful and correctly flags that the classical M-estimation theorems do not apply automatically. The description of ν^*, however, asserts an optimality that is not proved and is not true.

Minimising the expected action under the data-generating law is one objective. The objectives that matter for a reweighting object are others: the chi-square divergence, the SNIS asymptotic variance, the total-variation truth error, and the combined risk $\eta^2_{\nu,M} + \sigma^2(\nu)/N$ that §11 itself writes down. Because $Q_{\nu,M}$ depends on Π_M and λ as well as on ν, there is no reason for the minimisers to coincide, and in the paper's own example they do not. Sweeping the surrogate's restoring rate over the worked example's proposal and taking the observed law to be that proposal, I find

$$\nu^* :\ k = 2.75, \qquad \nu^\dagger := \arg\min_\nu \sigma^2(\nu) :\ k = 12.36,$$

with $\sigma^2 = 20.4$ at ν^* against 4.999 at $\nu^\dagger$ — a factor of 4.08 in asymptotic variance separating the two. (Both improve on the $k_\nu = 2$ actually used, whose $\sigma^2 = 30.89$.)

The manuscript in fact defines $\nu^\dagger$ itself, as the first of its open problems in §11, without remarking that it is a different object from the ν^* of §8.4. The two should be introduced together and distinguished: ν^* is the action-risk minimiser, and reserving the word "optimal" for the variance, divergence or risk objectives would remove the clash. **[R]**

3.6 The worked example (§9)

Two observations arise from the reimplementation.

Finding 6 (The example does not exercise the Itô term)**.** The divergence sum in (7) is what distinguishes the OM action from the Girsanov energy. For the Ornstein–Uhlenbeck surrogate $\nabla \cdot b_\nu = -k_\nu d$ is constant, so the divergence sum contributes the constant $-k_\nu T/2$ to the action and the factor $e^{\lambda k_\nu T/2}$ to every weight (which cancels identically in the self-normalised ratio (10)). The one worked example therefore leaves the OM-specific term untested: every number in §9 would be unchanged if the divergence sum were deleted from the definition.

This matters because it is precisely for the surrogates where the term is not constant (the polynomial drifts of §8.2, where the divergence is unbounded below and Tier II is unavailable) that the theory's harder cases live. A second example with a non-constant divergence, even a scalar double well, would exercise the term, would test Tier I in the regime for which it was written, and would show whether the Itô correction materially reshapes the selected measure. **[R]**

Finding 7 (The narrative attached to Table 1 is not estimable at fifty replications)**.** Table 1 reports $\sqrt{N} \cdot \text{std}$ over 50 replications and reads the column as rising toward the exact asymptote $\sqrt{\sigma^2(\nu)} \approx 5.56$ while remaining "$\approx 14\%$ below it at $N = 10^5$", attributing the gap to negative bias from rare large weights.

The direction is right; the figure is not supported. I ran ten *independent* batches of fifty replications at $N = 10^4$ and obtained $\sqrt{N} \cdot \text{std}$ values from 3.30 to 4.57, a spread of 39% across batches of the size used, against the single value 4.75 printed. A separate run of my own at fifty replications produced 7.53 at the same N, above the asymptote. Pooling 700 replications gives 3.44. The statistic's own sampling error at fifty replications is therefore larger than the discrepancy being interpreted, and the non-monotonicity the caption acknowledges is the same phenomenon.

The underlying claim survives independently: pooling, $\sqrt{N} \cdot \text{std} = 3.44$ at $N = 10^4$ against an asymptote of 5.56 is a genuine and substantial negative bias. The remedy is to raise the replication count and report a spread, not to soften the claim. The same applies to the single plug-in figure $\hat{\chi}^2 \approx 623$ at $N = 2 \times 10^5$: my independent draw at the same sample size gives 833.6 against the exact 1372.4, so the severe negative bias is confirmed while the particular number is one draw of a highly variable statistic.

Two supporting measures should accompany any retention of the table as evidence rather than illustration. Neither the random seeds nor the code are archived, so the table is not exactly reproducible even in principle; and in a regime this heavy-tailed the closed forms, which are exactly reproducible, should be allowed to carry the evidentiary weight, with the simulation presented as illustration. **[R]**

3.7 Structural comparison and open problems (§10–§11)

The comparison table is accurate in each cell, and the two surviving distinctions are correctly identified as the corrected exclusion (E1) and its likelihood-level consequence. The withdrawal in §10.4 of a third claimed distinction is correct.

Grid dependence deserves a subsection of its own. The manuscript's position that the grid is the model and not a truncation is defensible, and it is stated plainly. But the consequences are not drawn. Nothing establishes invariance of $Q_{\nu,M}$ under the discretisation convention (the prepoint Euler rule of (7) against a midpoint rule), under coordinate reparameterisation, or across nested grids; $F^*(\nu)$ changes when any of these changes, so M, λ, the proposal and the discretisation convention are all part of the definition of the estimand rather than numerical settings. Any empirical use must therefore report sensitivity to all four. **[R]**

The proposed continuum route does not survive testing. Of the five open problems, §11 proposes for the second a temperature schedule $\lambda_M \downarrow$ calibrated to the blow-up quantified in §9.4. I tested that route directly. Solving for the schedule that holds the chi-square divergence at a fixed value and measuring what selection remains (the displacement of the estimand from its proposal value) gives

	holding $\chi^2 = 10$		holding $\chi^2 = 1372.4$	
M	λ_M	selection displacement	λ_M	selection displacement
50	0.4307	0.1903	1.0000	0.2800
100	0.2750	0.1438	0.5777	0.2210
200	0.1815	0.1069	0.3581	0.1697
400	0.1224	0.0784	0.2318	0.1275
800	0.0837	0.0569	0.1544	0.0943

In both columns the displacement decays toward zero. By (1) the selection strength and the singularity are controlled by the same parameter, so a schedule that keeps the estimator usable drives the construction toward the proposal itself. This is evidence against the proposed route, not a theorem; but it is evidence obtained inside the manuscript's own example, and it suggests that a continuum theory requires a different renormalisation (subtracting a reference kinetic term, which is exactly the operation that converts the selection weight into a density ratio).

4 Originality: what is new and what is standard

Standard, and correctly attributed. The classical OM action and its tube-density and rate-functional roles; the SNIS estimator and its asymptotic variance; Chebyshev, Cantelli, Bernstein and Hoeffding; the chi-square characterisation of effective sample size; the Chatterjee–Diaconis and Sanz-Alonso sample-size scales; the Bissiri–Holmes–Walker generalised posterior; Mori–Zwanzig projection; White's misspecification theory. Each one is appropriately cited at the point of use.

The component theorems should not carry the originality claim. Theorems A, B and C are standard self-normalised importance-sampling and concentration theory specialised to the chosen loss. Gibbs reweighting by $e^{-\lambda \ell}$ is the classical generalised-Bayes update; the chi-square and Kullback–Leibler degeneracy diagnostics are mature; the Gaussian worked example is pedagogically useful but not itself new. The abstract and the contribution list present these in the same register as the material that is the paper's own, and should separate the two.

The paper's own. The assembly is the contribution: a coherent estimation theory for a setting in which the generator-based apparatus is unavailable, with the estimand stated exactly and the truth link isolated as a hypothesis rather than smuggled into a theorem. Within that, three items are genuinely the paper's: the finite-grid formulation adopted on the strength of a proved degeneracy rather than for convenience; the two-tier concentration structure matched to the two weight regimes that actually occur, together with the observation that its own Tier II is vacuous in its own example; and the pairing of a conditional transfer with a kernel-level obstruction, which makes the shape of the remaining gap explicit rather than rhetorical (once the inference corrected in Finding 3 is removed).

What is new but unclaimed. The multiplicative use of a path density, as opposed to the ratio used throughout the reweighting literature, is the construction's actual point of departure (Finding 2). The paper does not identify it, and claims novelty instead for a "third operational role" that is harder to defend (Finding 8).

Level. Moderate as an integrated methodological synthesis; low to moderate for the individual statistical results. The theorems are applications of known inequalities, carefully matched to hypotheses; the originality lies in the framing and the scoping rather than in the analysis.

5 Priority in the literature

Finding 8 (The third operational role is not positioned against path reweighting)**.** The framing claim is that the action is developed "in a third role", distinct from the tube-density and rate-functional roles: as a functional acting on separately supplied candidate paths. Reweighting an externally generated path ensemble by a path-space action is established practice, and two lines of it are not cited.

Donati, Hartmann and Keller reweight path ensembles generated under a reference potential to obtain path-ensemble averages under a perturbed one, using the Girsanov change of measure on path space; the paths are produced by a separate simulation and then reweighted, which is structurally the operation the manuscript describes. Adib shows that the two standard discretised path actions (one with and one without the second-derivative term, the same distinction as the divergence sum in (7)) are equivalent for the purposes of reweighting and sampling trajectories, and carries out the reweighting of an ensemble generated under one dynamics to another.

The manuscript does cite neighbouring work: the path-space variational coarse-graining of Harmandaris and coauthors, which it correctly identifies as the closest published prior art, and the control-based importance sampling of Hartmann and Schütte and of Zhang and coauthors. But those are all *optimisation* constructions, and the two distinctions drawn in §10.2 (that they presuppose a path-space relative entropy, and that they optimise the surrogate toward the truth) do not apply to Donati and coauthors or to Adib, who optimise nothing and simply reweight.

The distinction that does survive against this literature is sharper than the one claimed, and it is the one identified in Finding 2: every reweighting method listed above uses a *difference* of actions, equivalently a ratio of path densities with a common diffusion coefficient, which is why their weights are likelihood ratios and their chi-square divergences stay bounded under refinement. The manuscript uses a *single* action, equivalently a product of two path densities. That is a genuine departure, it is the source of everything distinctive about the construction's behaviour, and it should carry the novelty claim in place of the three-roles framing. **[M]**

6 Specific criticisms

Items are marked **[M]** where they bear on the correctness or the substance of a claim, and **[R]** where they concern presentation, attribution or scope.

[M] The inference opening §7.2 that the conditional structure is "forced" is invalid (Finding 3), and contradicts the same section four paragraphs later. Kernel-norm separation of a divisible surrogate and total-variation separation of the selected measure are logically independent; an explicit construction with zero selection bias and an indivisible target is recorded in the verification section.

[M] The selection weight is the surrogate's own path density and the selected measure is a product of densities
(Finding 2). The identification is exact for constant-divergence surrogates, it fixes the selected measure's diffusion coefficient through (1), and it is the mechanism behind the refinement degeneracy the paper reports but attributes elsewhere.

[M] For proposals that are not designed against the tilt, the transfer hypothesis fails under refinement
(Finding 4). The total-variation separation between the selected measure and the proposal rises to 0.99999 at $M = 400$. No invariant of the target family can control a quantity that belongs to the proposal–surrogate pair; the open problem of Remark 7.2 should be restated as a design problem.

[M] The third-role novelty claim is not positioned against the path-reweighting literature
(Finding 8). The surviving and stronger distinction is the use of a single action rather than a difference.

[R] The reading placed on Proposition 2.3 is too broad
(Finding 1); what is degenerate is the pointwise weight (6), not continuous-path inference, as the bounded chi-square of the ratio weight on the same grids shows.

[R] The calibrated parameter is called the best reweighting parameter
(Finding 5); it is the action-risk minimiser, and differs from the variance minimiser by a factor of four in the paper's own example.

[R] The worked example does not exercise the Itô-correction term
(Finding 6); for the Ornstein–Uhlenbeck surrogate it cancels identically from the estimator. A non-constant-divergence example is needed.

[R] The variance narrative of Table 1 is not estimable at fifty replications
(Finding 7); the underlying negative bias is real and survives pooling, and neither seeds nor code are archived.

[R] The divergence term is misattributed to the Itô form of the Girsanov derivative, which does not display it; it enters through the small-ball passage, and through Itô's formula only in the gradient-drift case.

[R] The effective-sample-size thresholds of Remark 6.2 are heuristics
presented in the register of the theorems; the smoothing method invoked carries its own shape diagnostic, which should be reported with them.

[R] Theorem 4.1(iii) is conditional on an unverified uniform integrability
for which no primitive sufficient condition is supplied.

[R] Grid and discretisation dependence is not addressed.
No invariance is established under the prepoint-versus-midpoint convention, under coordinate reparameterisation, or across nested grids, so M, λ, the proposal and the convention are all part of the estimand.

[R] The temperature schedule proposed for the continuum limit does not survive testing
in the paper's own example; the displacement of the estimand decays to zero along any schedule holding the chi-square divergence fixed.

[R] The estimand is not motivated.
The paper is candid that $F^*(\nu)$ depends on the proposal, the surrogate and the temperature and is not the target's expectation. It does not say who would want that number, or in what decision it is used, when the transfer hypothesis is unavailable.

7 Questions to the author

1. Show that $e^{-\lambda S_{\nu,M}}$ is proportional to the surrogate's Euler path density raised to λ for constant-divergence surrogates, and state the consequence $Q_{\nu,M} \propto \Pi_M\, p_\nu^\lambda$. Why multiply the two densities rather than divide, which is what every reweighting method in the adjacent literature does?

2. Derive (1) for the Gaussian case and say what it implies for the quadratic variation of the selected measure relative to the proposal.

3. Construct an explicit example in which a divisible surrogate family is bounded away from an indivisible target in kernel norm while the selected measure equals the true grid law exactly. What does that show about the sentence opening §7.2, and what remains of the claim that the conditional structure is forced?

4. Given (1), for which proposals is hypothesis (D1) attainable at all? Is the tolerance $\eta_{\nu,M}$ controlled by any property of the target, or only by the proposal–surrogate pair?

5. Is there a temperature schedule λ_M under which the continuum limit is both non-degenerate and non-trivial? The evidence from your own example is that holding the chi-square divergence fixed drives the selection displacement to zero.

6. Precisely which continuous-path object fails in Proposition 2.3, and why does the argument not rule out a Girsanov change of path measure or an Onsager–Machlup small-ball functional?

7. Derive the divergence term of the classical action from a small-ball or Stratonovich formulation. In what sense is it, and in what sense is it not, an Itô correction?

8. Position the construction against Girsanov path reweighting of externally generated ensembles and against reweighting with discretised stochastic actions. The two distinctions offered in §10.2 apply to optimisation methods and not to these.

9. Which results in Theorems A–C are new, and which are specialisations of established self-normalised importance-sampling and concentration theory?

10. Give primitive sufficient conditions for the uniform integrability assumed in Theorem 4.1(iii), or say that none is available.

11. Why does minimising the empirical surrogate action not minimise the asymptotic variance, the chi-square divergence or the truth-link error? Your own example separates the two minimisers by a factor of four in variance.

12. What exactly is estimated when M, λ, the proposal, or the discretisation convention changes? Is any estimand invariant across nested grids, and can the family $\{Q_{\nu,M}\}$ be made projectively consistent?

13. Supply a second worked example with non-constant drift divergence, so that the Itô term is exercised and Tier I is tested in the regime for which it was written.

14. Raise the replication count of the convergence study, archive the seeds and code, and report the spread of the variance statistic; at fifty replications its batch-to-batch variability exceeds the effect described in the caption.

15. How would the theory change for a state-dependent diffusion $\sigma(x)$, where the classical action is sensitive to the coordinate system and to the discretisation convention?

16. What empirical procedure would allow (D1) to be tested or bounded in an application where the true path law is unknown, and what decision consumes $F^*(\nu)$ when it cannot be?

8 Technical conclusion

The reproducibility standard is met without qualification. Every theorem presented are correct according to this review. The scoping is careful throughout, and in several places (the separation of the three exclusions, the refusal of a false finite-sample identity, the acknowledgement that the paper's own Tier II bounds are vacuous in its own example, the withdrawal of an unconditional bias bound with an explanation of why it must fail, and the statement of two published sample-size results as class statements rather than per-observable necessary conditions) the manuscript is more exacting about its own claims than the surrounding literature is about the same material.

Three defects bear on the substance, and they are connected. The sentence opening §7.2 (that the conditional structure of the transfer theorem is forced because no divisible surrogate drives the selection bias to zero) does not follow, and the same section contradicts it four paragraphs later. The selected measure is a tilt of the proposal, not the law of the surrogate, and I record a construction in which the selected measure equals an indivisible target's grid law exactly while the kernel obstruction applies in full. The reason the proposal can do this is the second defect: the selection weight is, up to a constant, the surrogate's own path density, so the selected measure is a product of two path densities rather than a ratio, and multiplying densities adds precisions. That single identity fixes the diffusion coefficient of the selected measure, explains the chi-square blow-up under refinement, states exactly which proposals make the transfer hypothesis attainable and which make it unattainable, and shows why the temperature schedule proposed for a continuum theory drives the selection to zero. It is nowhere in the manuscript. And the novelty claim should be repositioned on the same identity: reweighting an externally generated path ensemble by a path action is established practice, and the literature that does it is not cited; what distinguishes this construction is that it uses a single action where that literature uses a difference.

The remaining items concern scope and presentation. The reading placed on the continuous-path degeneracy is broader than the proposition supports, and a ratio weight on the same grids has bounded chi-square divergence, which shows that continuous-path reweighting is not what fails. The calibrated parameter is called optimal for an objective it does not optimise, and differs from the variance minimiser by a factor of four in the paper's own example. The divergence term is misattributed to the Itô form of the Girsanov derivative. The effective-sample-size cutoffs are heuristics presented in the register of the theorems. Grid and discretisation dependence is left unaddressed although it is part of the definition of the estimand. The one worked example is

Ornstein–Uhlenbeck, so the Itô-correction term cancels identically and the OM-specific part of the functional is never exercised. And the variance narrative attached to the convergence table is not estimable at fifty replications, although the negative bias it describes is real and survives pooling.